

```
Python 3.14.0b1 (tags/v3.14.0b1:b092705, May 7 2025, 10:22:31) [MSC v.1943 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
```

```
>>> # Classify computer vision level
```

```
...
... application = input("Enter application: ")
...
... if application.lower() == "surveillance":
...     print("Level: High-Level Computer Vision")
...     print("Reason: Detects, tracks and recognizes objects in real time.")
... else:
...     print("Application not found.")
...
...
```

```
Enter application: surveillance
```

```
Level: High-Level Computer Vision
```

```
Reason: Detects, tracks and recognizes objects in real time.
```

```
>>> |
```

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```
>>> # Compare image quality
```

```
...  
... pixel = int(input("Enter pixel resolution (MP): "))  
... intensity = int(input("Enter intensity resolution (bits): "))  
...  
... print("Pixel Resolution:", pixel, "MP")  
... print("Intensity Resolution:", intensity, "bits")  
...  
... if pixel >= 12:  
...     print("High spatial detail.")  
... else:  
...     print("Low spatial detail.")  
...  
... if intensity >= 8:  
...     print("Good brightness levels.")  
... else:  
...     print("Poor brightness levels.")  
...  
...
```

```
Enter pixel resolution (MP): 16
```

```
Enter intensity resolution (bits): 8
```

```
Pixel Resolution: 16 MP
```

```
Intensity Resolution: 8 bits
```

```
High spatial detail.
```

```
Good brightness levels.
```

```
>>> |
```

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```
>>> # Detect aliasing
```

```
... sampling = int(input("Enter sampling frequency: "))
... signal = int(input("Enter signal frequency: "))
...
... if sampling >= 2 * signal:
...     print("No aliasing.")
... else:
...     print("Aliasing detected.")
...     print("Correction: Increase sampling frequency or apply anti-aliasing filter.")
...
Enter sampling frequency: 50
Enter signal frequency: 40
Aliasing detected.
Correction: Increase sampling frequency or apply anti-aliasing filter.
>>> |
```



Python 3.14 (64-bit)



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```
>>> # Check image quality in low light
```

```
...
```

```
... light = int(input("Enter light intensity (0-100): "))
```

```
...
```

```
... if light >= 50:
```

```
...     print("Image captured with good brightness.")
```

```
... else:
```

```
...     print("Image is dark and noisy.")
```

```
...     print("Suggestion: Increase exposure or use better lighting.")
```

```
...
```

```
Enter light intensity (0-100): 30
```

```
Image is dark and noisy.
```

```
Suggestion: Increase exposure or use better lighting.
```

```
>>> |
```

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```
>>> # Check image quality based on sampling
```

```
... sampling = int(input("Enter sampling rate (pixels): "))
...
... if sampling >= 300:
...     print("Image Quality: Clear")
...     print("Correction: No correction needed")
... else:
...     print("Image Quality: Blurred")
...     print("Correction: Increase sampling rate or image resolution")
...
```

```
Enter sampling rate (pixels): 150
```

```
Image Quality: Blurred
```

```
Correction: Increase sampling rate or image resolution
```

```
>>>
```